

ARMA-GARCH & Trend following

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Abstract

This research paper explores alternative approaches to trend-following investment strategies. These strategies depend on capturing and forecasting market trends—an inherently difficult task due to volatility clustering and long-memory effects. The study investigates the use of ARMA-GARCH-type models applied after recovering the trend of an asset's price.

1 Introduction

In this paper, we address the problem of decomposing a stock price into two components: trend and volatility. The more information is captured by the trend, the less remains for the volatility, and vice versa. This trade-off forms the starting point of our work. The goal is not only to find a balance between these two components, but also to incorporate features such as long memory and volatility clustering.

We propose a modeling framework that integrates **FARIMA (Fractionally Integrated ARMA)** models with **GARCH (Generalized Autoregressive Conditional Heteroskedasticity)** processes. ARFIMA models capture long-range dependencies in financial time series, while GARCH models handle time-varying volatility. By applying these models sequentially, we aim to extract the structure of the data without ignoring the trend component of our financial series.

Our methodology consists in first isolating the trend and then analyzing the residuals—i.e., the difference between the original financial time series and the extracted trend. We then fit an ARMA-like model to these residuals. The difference between the residuals and the ARMA-fitted values is modeled using a GARCH-type model. The ultimate objective is to recover white noise, indicating that we have effectively captured the structure of the asset's dynamics.

This framework can serve as a supporting tool within trend-following strategies. It is designed to complement traditional indicators (e.g., MACD, DMI, moving averages), as it does not attempt to predict trend reversals.

The paper is organized as follows. Section 2 presents the methodology and the proposed model. Section 3 provides graphical illustrations of the main example and discussions about other assets.

2 Methodology and Theory

2.1 Models

2.1.1 ARMA-type models

The three models used in our study are ARMA, ARIMA and ARFIMA.

The **Autoregressive Moving Average** model, denoted as $\text{ARMA}(p, q)$, combines two components:

- An **autoregressive (AR)** part of order p , where the current value depends linearly on its past p values;
- A **moving average (MA)** part of order q , where the current value depends linearly on the past q white noise error terms.

The ARMA model is defined as:

$$X_t = \sum_{i=1}^p \phi_i X_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j},$$

where ε_t is a white noise process.

It was popularized by Box and Jenkins in the foundational text on time series forecasting [BJRL15].

The **Autoregressive Integrated Moving Average** model, denoted $\text{ARIMA}(p, d, q)$, generalizes ARMA by including an integration order d to account for non-stationary time series. The model applies d differencing operations to the original series to induce stationarity.

Formally, the differenced series $X_t = \Delta^d Y_t$ follows an $\text{ARMA}(p, q)$ process:

$$\Delta^d Y_t = \sum_{i=1}^p \phi_i \Delta^d Y_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}.$$

ARIMA models are widely used in macroeconomic and financial forecasting when the data exhibit trends or structural shifts [BJRL15].

The **Autoregressive Fractionally Integrated Moving Average** model, or $\text{ARFIMA}(p, d, q)$, allows the differencing parameter d to be a real (fractional) value. This flexibility enables the model to capture *long memory* behavior, where autocorrelations decay slowly and persist over time.

The fractional differencing operator is defined as:

$$(1 - B)^d X_t = \sum_{k=0}^{\infty} \binom{d}{k} (-1)^k X_{t-k},$$

where B is the backward shift operator, i.e., $BX_t = X_{t-1}$. When $0 < d < 0.5$, the process is stationary with long-range dependence.

This model was introduced by [Hos81] and has been further developed in econometric applications by [Bai96].

These models are particularly useful in financial econometrics for modeling volatility, returns, and persistent structures in economic time series

2.1.2 GARCH-type models

The two models used in our study are GARCH and EGARCH.

The **Autoregressive Conditional Heteroskedasticity** model, or **ARCH**, was introduced by [Eng82]. It models time-varying volatility by allowing the variance of the current error term to depend on the squared past error terms. The $\text{ARCH}(q)$ model is defined as:

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim \mathcal{N}(0, 1), \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2,$$

where $\alpha_0 > 0$ and $\alpha_i \geq 0$ for $i = 1, \dots, q$ to ensure positive variance.

The **Generalized ARCH** or **GARCH** model, introduced by [Bol86], generalizes ARCH by including lagged conditional variances. A $\text{GARCH}(p, q)$ model is given by:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2,$$

which allows for more persistent volatility modeling.

To account for asymmetries in volatility (such as the leverage effect), the **Exponential GARCH** or **EGARCH** model was proposed by [Nel91]. The $\text{EGARCH}(1, 1)$ model expresses the logarithm of the conditional variance:

$$\log(\sigma_t^2) = \omega + \beta \log(\sigma_{t-1}^2) + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \alpha \left(\left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| - \mathbb{E} \left[\left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| \right] \right).$$

This specification ensures positivity of the variance without imposing parameter constraints and can capture asymmetries between positive and negative shocks.

2.2 Methodology

Our methodology for analyzing time series and building the strategy follows a sequential approach. This structure has its limitations, particularly due to the dependency of each step on the previous one errors can therefore propagate throughout the entire process. However, this is a deliberate and meaningful choice to adopt the trend as the cornerstone of the analysis, even though alternative modeling paths could have been explored.

The first step involves detecting trend breakpoints using a spline-based algorithm from Python’s `scipy.interpolate` library [VGO⁺20]. Based on these breakpoints (or nodes), we fit piecewise polynomial regressions of degree 3, imposing a continuity constraint by ensuring that each segment begins where the previous one ends. This results in a sequence of cubic polynomials that are continuous, but not necessarily differentiable at the breakpoints.

Let the original time series be denoted by Y_t , and the estimated spline-based trend by $\hat{T}_t$. We define the **first-order residuals** as:

$$r_t^{(1)} = Y_t - \hat{T}_t,$$

representing the residual component after removing the trend. On these residuals $r_t^{(1)}$, we fit either an ARMA, ARIMA, or ARFIMA model, selecting the model with the lowest Akaike Information Criterion (AIC). In the case of an ARFIMA, we integrate R code using the `fractdiff` package [MWHL22] to estimate the fractional differencing parameter.

Next, we model the volatility movements by applying an ARCH, GARCH, or EGARCH model to the **second-order residuals**, defined as:

$$r_t^{(2)} = r_t^{(1)} - \hat{r}_t^{(1)},$$

where $\hat{r}_t^{(1)}$ refers to the fitted values from the selected ARMA-type model.

To evaluate the decomposition, we perform a Ljung–Box test on the standardized residuals from the volatility model. If neither the residuals nor the squared residuals exhibit autocorrelation, it suggests that our approach has successfully captured the underlying structure of the data.

If autocorrelation persists, we apply structural break detection on the conditional standard deviation using the `ruptures` package in Python [TOV20]. This procedure identifies change points, which may indicate a change of model over time. The method is then repeated on the resulting time sub-intervals, provided that adjusting the trend breakpoints does not improve the model.

Introducing multiple regimes and corresponding models often proves more effective than merely refining the trend, though it tends to hinder predictive robustness.

Although this evaluation method is biased and does not guarantee predictive accuracy, it allows us to fully implement and explore the logic of our proposed approach.

3 Examples

3.1 Gold

Our main example focuses on front-month COMEX gold futures over the period from January 01, 2015, to May 01, 2025. This dataset was selected because it spans several distinct economic states, including the pre-COVID period, the COVID crisis, the post-COVID recovery, and the “Trump era.”

Figure 1 displays the raw time series along with the modeled trend over the entire sample period. Notably, this dataset does not require any trend breakpoints to ultimately recover white noise, suggesting a relatively stable underlying structure and a proximity of the data points to long-term trends.

The first-order residuals $r_t^{(1)}$ are shown in Annex Figure 5. Based on the Akaike Information Criterion (AIC), an ARIMA(2,2,4) model is selected for these residuals (see Annex, Figure 6). Suggestion of heteroskedasticity in the second-order residuals $r_t^{(2)}$ is visible in Figure 2.

Although the main example does not rely on an ARFIMA model for this specific time window, we observe that ARFIMA models can perform marginally better in other time periods or with different assets. These improvements are generally too minor to justify the added complexity. While disappointing, the ARFIMA model still provides a useful comparison and may serve as an improvement for future developments.

A GARCH-type model is fitted to the second-order residuals, resulting in an EGARCH(2,1) model with a standardized skewed Student’s t-distribution (see Annex, Figure 7). The histogram of the standardized residuals illustrates the goodness-of-fit to this distribution (Figure 3). The plot of the conditional standard deviation (Figure 4) highlights the heteroskedasticity previously hinted at by the second-order residuals.

Ultimately, we recover white noise, indicating that all exploitable structure has been extracted from the data.

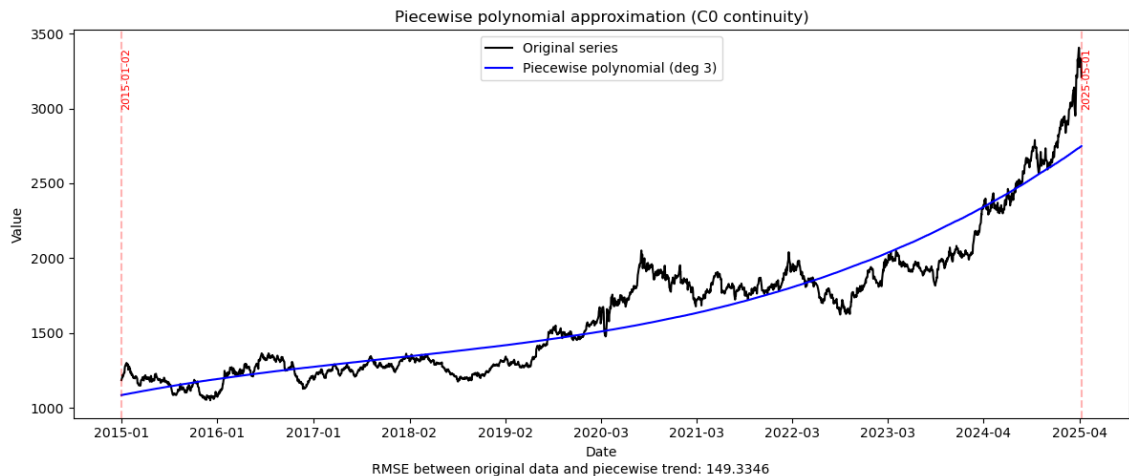


Figure 1:

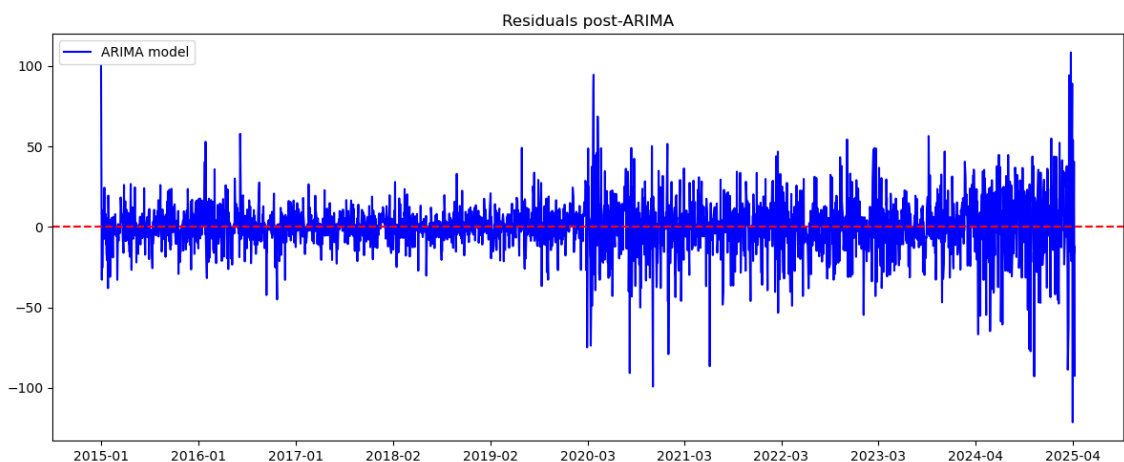


Figure 2:

As to the prediction, no evaluation metric was found or considered sufficient to definitively validate or reject the proposed approach. Nevertheless, the method yields intriguing results (see Figures 8 and 9).

3.2 Discussion on other assets

We also applied this method to other assets, where it successfully captured the underlying structure. Notable examples include LVMH (Annex Figures 10 and 11) and Wolters Kluwer.

The multiplication of trend segments tends to harm long-term predictions, as trends are likely to reverse. However, the LVMH example highlights the importance of adapting the trend length: introducing an additional trend breakpoint can improve short-term forecasting performance.

Conversely, there are assets on which the method performs poorly, either due to strong exogenous factors or lack of flexibility in the model. For instance, in the cases of Nvidia and Tesla, high volatility and extreme sensitivity prevent stationarity. This is due to the essence of our work, since it was built to support trend-following strategies, not to directly forecast a share price from its history.

4 Conclusion

In conclusion, we introduced a new framework for modeling trends by decomposing an asset's price into two sequential components: trend and volatility. The choice of a sequential approach reflects a deliberate bias toward emphasizing the trend as the primary structural element.

Though the method opens up promising opportunities, it currently lacks extensive benchmarking and predictive performance metrics. Its applicability also varies across assets, performing best

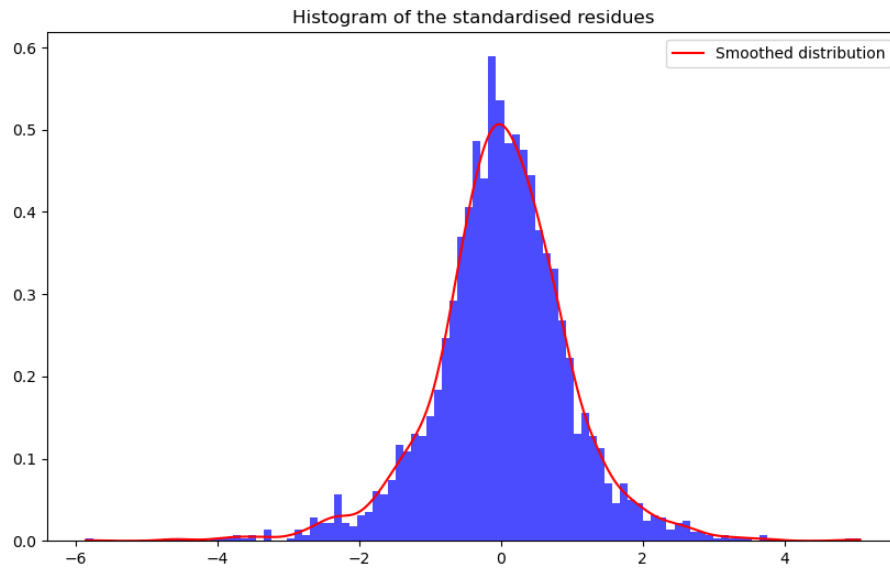


Figure 3: The histogram of the standardized residues of the EGARCH

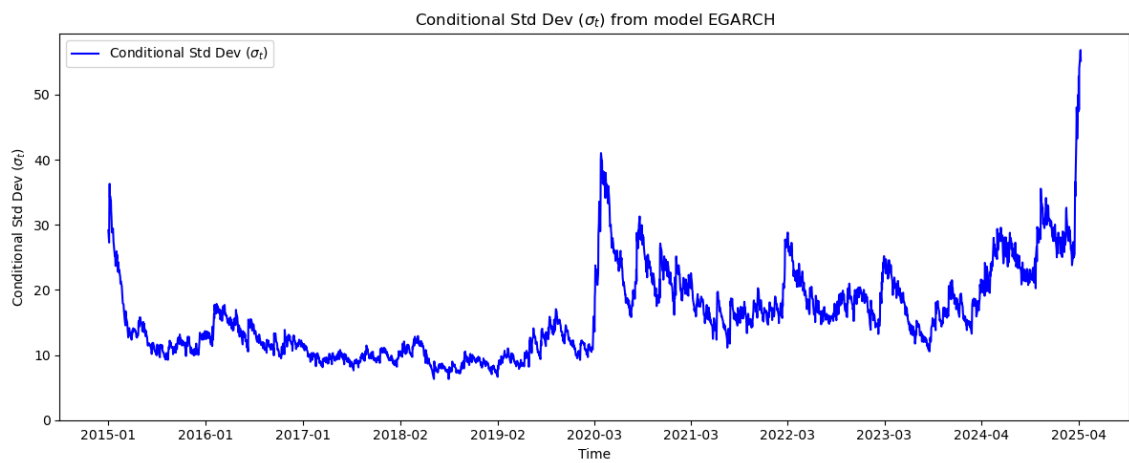


Figure 4: The conditional standard deviation resulting from the EGARCH showing clusters of volatility

on those exhibiting long and persistent trends. Despite these limitations, the approach provides a valuable perspective for trend-following strategies and offers a solid foundation for further exploration, such as applying the framework to high-frequency data, including cryptocurrency prices with second-level granularity and half-hour trends.

A Annexes

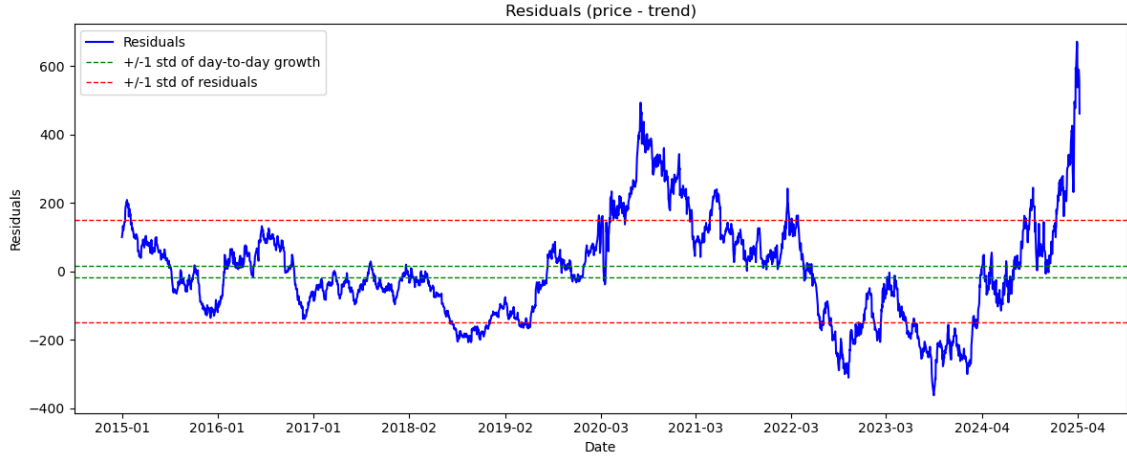


Figure 5: These are the residuals between the price of gold and the trend

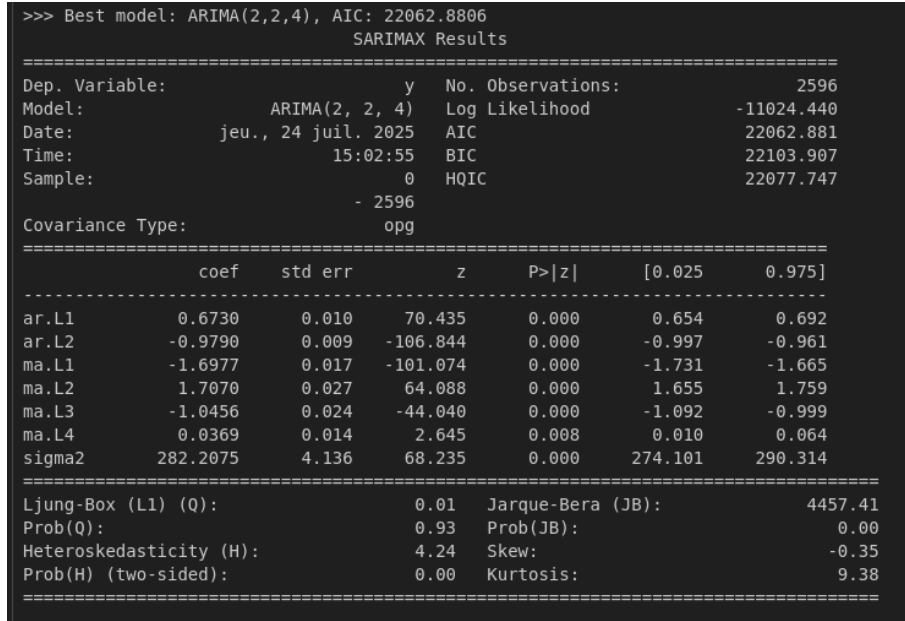


Figure 6: Those are the parameters of the ARIMA modeled on the residuals $r_t^{(1)}$

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>>> Best model: EGARCH with skewt, p=2, q=1, AIC: 21014.2074
('EGARCH', 'skewt', 2, 1, 21014.20737195001,
Zero Mean - EGARCH Model Results
=====
Dep. Variable:          y      R-squared:          0.000
Mean Model:            Zero Mean  Adj. R-squared:      0.000
Vol Model:             EGARCH    Log-Likelihood:    -10501.1
Distribution:          Standardized Skew Student's t  AIC:          21014.2
Method:               Maximum Likelihood             BIC:          21049.4
Date:                 jeu., juil. 24 2025             No. Observations:  2596
Time:                 15:02:58                       Df Residuals:      2596
                                           Df Model:          0
Volatility Model
=====
              coef    std err          t      P>|t|      95.0% Conf. Int.
-----
omega         0.0340    1.505e-02     2.262   2.370e-02   [4.545e-03, 6.352e-02]
alpha[1]      -0.0529    6.017e-02    -0.879   0.379   [-0.171, 6.504e-02]
alpha[2]       0.1780    6.291e-02     2.830   4.657e-03   [5.473e-02, 0.301]
beta[1]        0.9959    2.619e-03    380.212   0.000   [ 0.991, 1.001]
Distribution
=====
              coef    std err          t      P>|t|      95.0% Conf. Int.
-----
eta           4.1677     0.333    12.529   5.192e-36   [ 3.516, 4.820]
lambda       -0.0486    2.263e-02    -2.148   3.171e-02   [-9.297e-02, -4.256e-03]
=====

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Figure 7: Those are the parameters of the EGARCH modeled on the residuals $r_t^{(2)}$

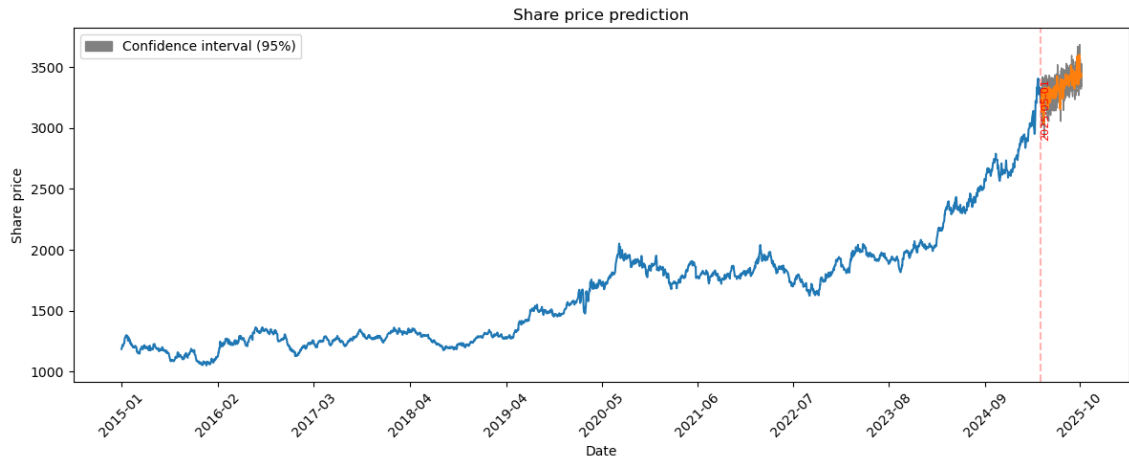


Figure 8: The prediction of the future values of the futures on gold

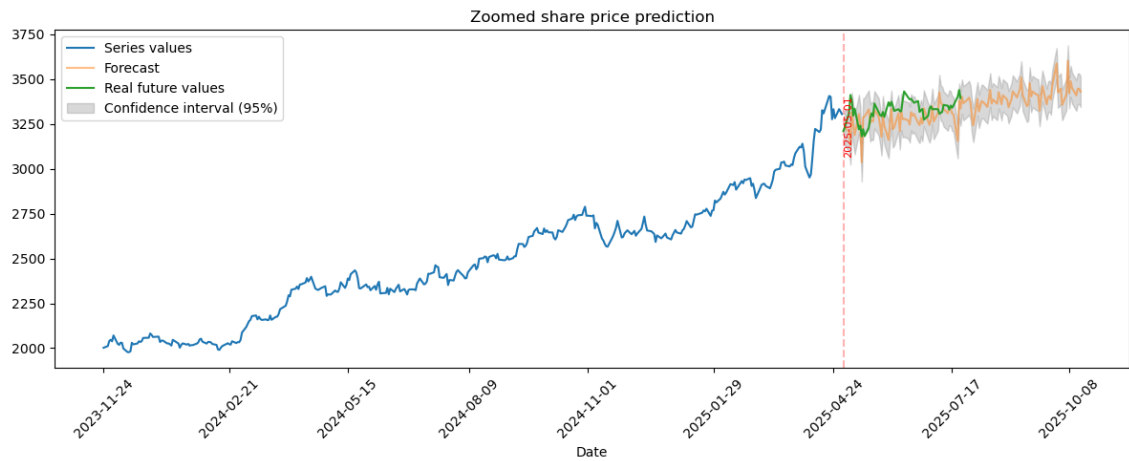


Figure 9: The zoomed prediction of the future values of the futures on gold

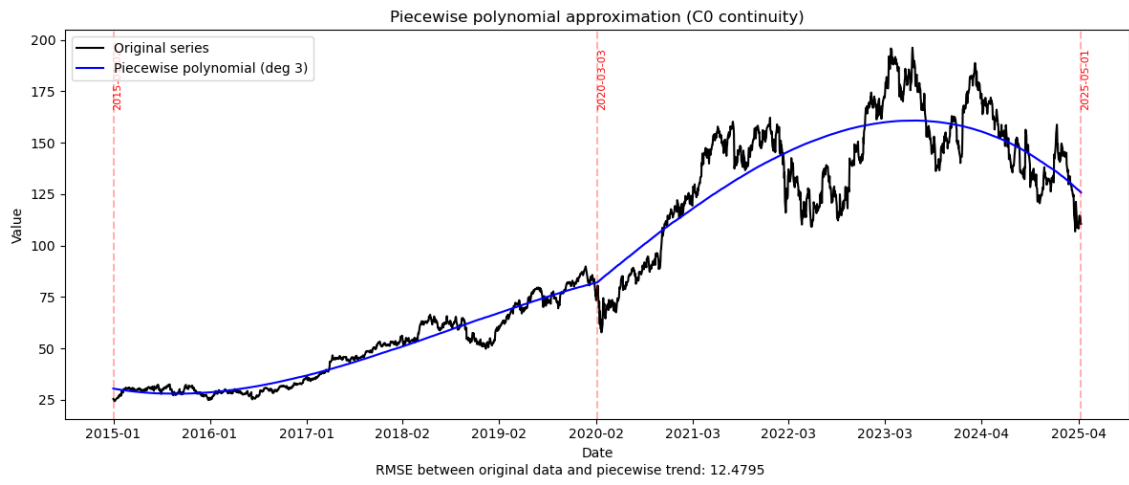


Figure 10: Trend approximation of LVMH share price

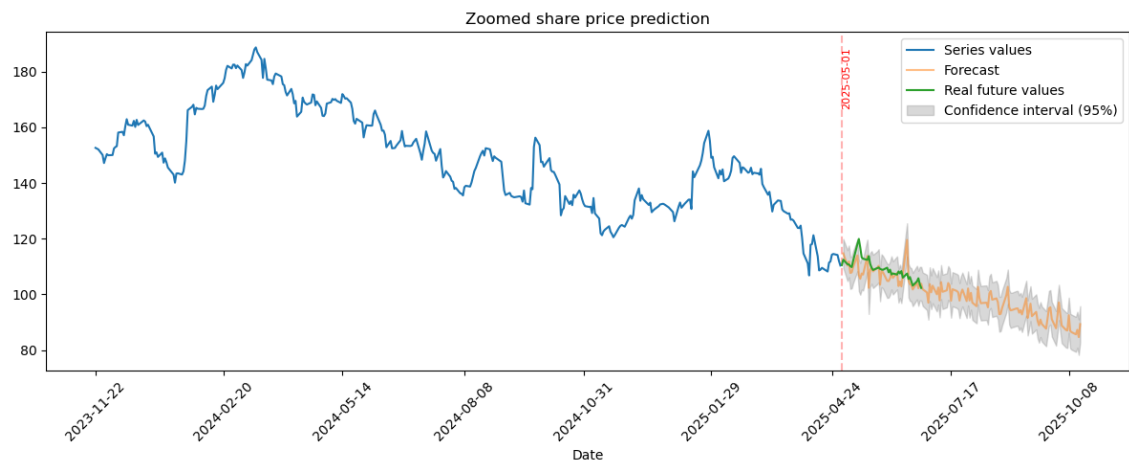


Figure 11: Prediction for LVMH share price

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